

Math Notes

Sunday, June 7, 2026 5:21 PM

Statement: A statement is a declarative sentence which is either true or false but not both simultaneously.

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Statements are denoted by $p, q, r, \dots$

Truth value

True 'T'

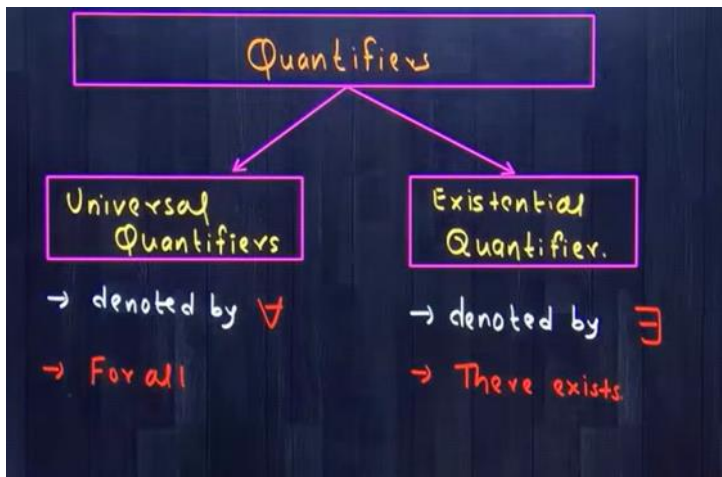
False 'F'

Example 3.4.4 Express following statement in propositional form :

- There are many clouds in the sky but it did not rain.
- I will get first class if and only if I study well and score above 80 in mathematics.
- Computers are cheap but softwares are costly.
- It is very hot and humid or Ramesh is having heart problem.
- In small restaurants the food is good and service is poor.

Associativity of $\wedge$	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
Distributivity of $\wedge$ over $\vee$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Distributivity of $\vee$ over $\wedge$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Double negation	$p \equiv \sim(\sim p)$
De Morgan's laws	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
De Morgan's laws	$\sim(p \wedge q) \equiv \sim p \vee \sim q$
Tautology	$p \vee \sim p \equiv \text{Tautology}$
Contradiction	$p \wedge \sim p \equiv \text{Contradiction}$
Absorption laws	$p \vee (p \wedge q) \equiv p$
Absorption laws	$p \wedge (p \vee q) \equiv p$

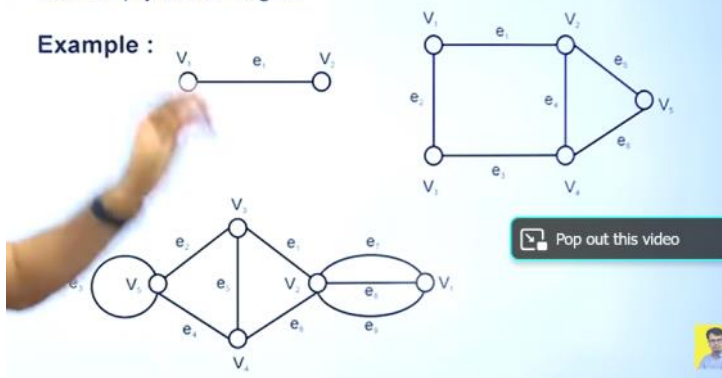
Idempotence of $\vee$	$p \equiv p \vee p$
Idempotence of $\wedge$	$p \equiv p \wedge p$
Commutativity of $\vee$	$p \vee q \equiv q \vee p$
Commutativity of $\wedge$	$p \wedge q \equiv q \wedge p$
Associativity of $\vee$	$p \vee (q \vee r) \equiv (p \vee q) \vee r$



GRAPH

A graph G is mathematical structure consisting of two sets V and E where V is a non-empty set of vertices and E is a non-empty set of edges.

Example :



BASIC TERMINOLOGY

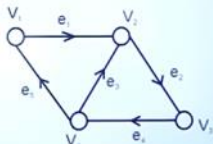
(1) **Trivial Graph** : A graph consisting only one vertex and no edge.

Example : 

(2) **Null Graph** : A graph consisting n vertices and no edge.

Example : 

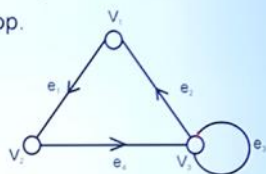
(3) **Directed Graph** : A graph consist the direction of edges then this is called directed graph.



(4) **Undirected Graph** : A graph which is not directed then it is called undirected graph.

(5) **Self loop in a Graph** : If edge having the same vertex as both its end vertices is called self loop.

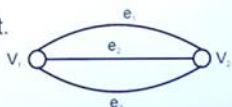
Example :



(6) **Proper edge** : An edge which is not self loop is called proper edge.

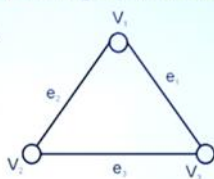
(7) **Multi edge** : A collection of two or more edges having identically end point.

Example :



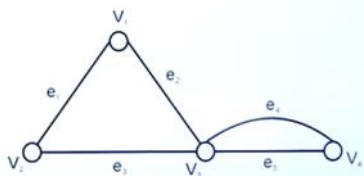
e_1, e_2, e_3 are multiedge.

(8) **Simple Graph** : A graph does not contain any self loop and multiedge.



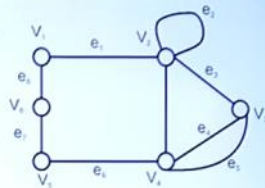
(9) **Multigraph** : A graph does not contain any self loop but contain multiedge is called multigraph.

Example :



(10) Pseudo Graph : A graph contain both self loop and multiedge is called pseudo graph.

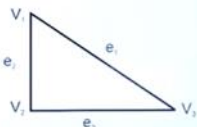
Example :



(11) Incidence and Adjacency : Let e_k be an edge joining two vertices V_i & V_j then e_k is said to be incident of V_i & V_j .

Two vertices are said to be adjacent if there exist an edge joining this vertices.

Example :



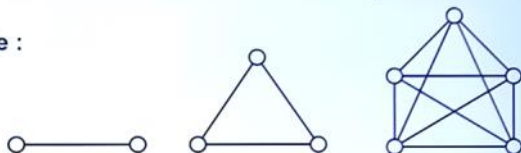
Here e_1 is incident of V_1 & V_2 and V_1 & V_2 are adjacent but

V_2 & V_3 are not adjacent

SOME IMPORTANT GRAPH

(1) Complete Graph : A simple connected graph is said to be complete if each vertex is connected to every other vertex.

Example :



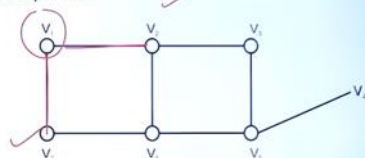
(2) Regular Graph : A graph G is said to be regular if every vertex has the same degree. If degree of each vertex of graph G is K , then it is called k -regular graph.

Example :



(3) Bigraph (or Bipartite) : If the vertex set V of a graph G can be partitioned into two non-empty disjoint subsets X and Y in such a way that edge of G has one end in X and one end in Y . Then G is called bipartite.

Example :



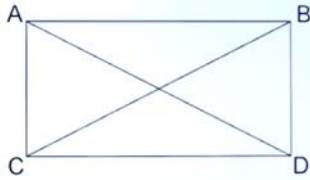
Here $V = \{V_1, V_2, V_3, V_4, V_5, V_6, V_7\}$

$X = \{V_1, V_2, V_3, V_4\}$ and $Y = \{V_5, V_6, V_7\}$

CONNECTED GRAPH

An undirected graph is said to be connected if there is a path between every two vertices.

Example :

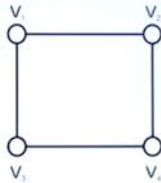


Note : If a graph is connected then it will not bipartite.

(4) **Complete Bipartite Graph** : If every vertex in X is disjoint from every vertex in Y , then it is called a complete bipartite graph.

If X and Y contain m & n vertices then this graph is denoted by $K_{m,n}$.

Example :



Here $X = \{v_1, v_2\}$

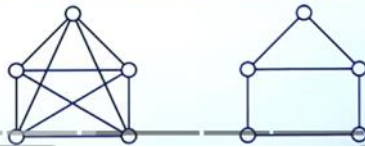
& $Y = \{v_3, v_4\}$

It is denoted by $K_{2,2}$

(5) **Subgraph** : Let $G(V, E)$ be a graph. Let V' be a subset of V and let E' be a subset of E whose end point belong to V' .

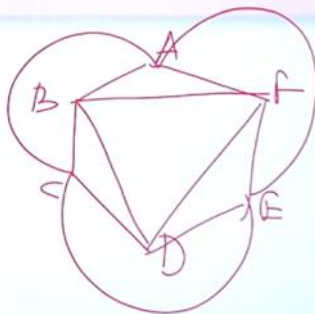
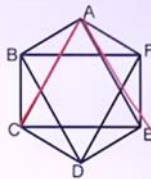
Then $G(V', E')$ is a graph and called a subgraph of $G(V, E)$.

Example :



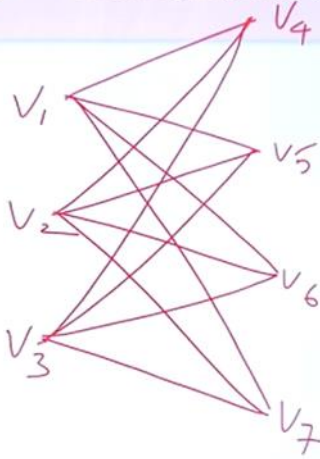
1

Draw a planar representation of a graph.



2

Draw $K_{3,4}$ bipartite graph.



5

HANDSHAKING THEOREM

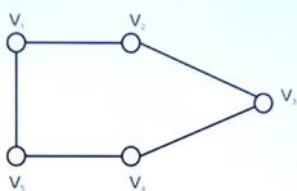
The sum of the degree of the vertices of a graph G is equal to twice the number of edges in G .

Proof : Consider a graph G with ' e ' edges and ' n ' vertices $V_1, V_2, \dots, V_n$. Since each edge contributes 2 degree in a graph the sum of the degree of all vertices will be a

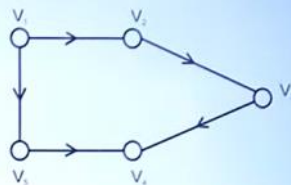
$$\sum_{i=1}^n \deg(v_i) = 2e$$

$$\Rightarrow \sum_{i=1}^n \deg(v_i) = 2 \times \text{Number of edges}$$

Example :



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$



$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

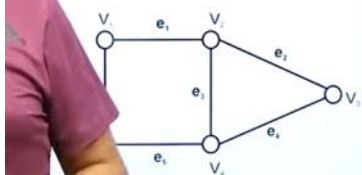
(2) **Incidence Matrix** : Let G be a graph with m vertices

$V_1, V_2, \dots, V_m$ and n edges $e_1, e_2, \dots, e_n$.

Let a matrix $M = [m_{ij}]_{m \times n}$ defined by

$$m_{ij} = \begin{cases} 1 & \text{if the vertex } V_i \text{ is incident on the } e_j \\ 0 & \text{if } V_i \text{ is not incident on } e_j \\ 2 & \text{if } V_i \text{ is an end of the loop } e_j \end{cases}$$

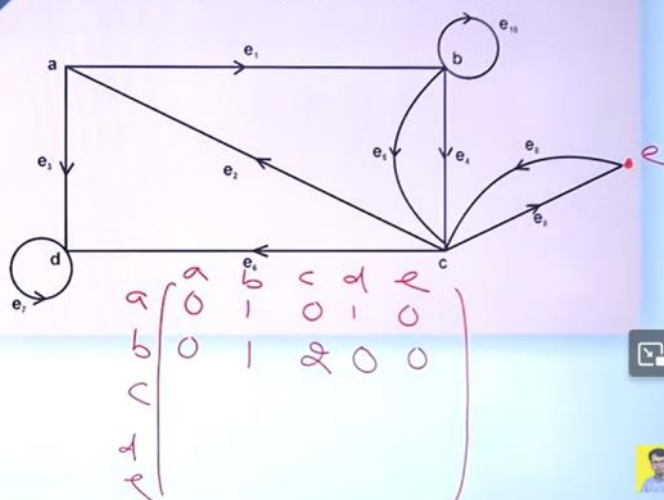
Example :



$$A = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

2

Find the Adjacency matrix of graph.



ISOMORPHISM OF GRAPH

Two graphs G_1 and G_2 are said to be isomorphic if there is one to one correspondence between the edge set E_1 & E_2 in such a way that if e_1 is an edge with end vertices u_1 & v_1 in G_1 , then the corresponding edge e_2 in G_2 has its end point vertices u_2 & v_2 in G_2 which correspond to u_1 & v_1 .

Another Definition : Two graphs G_1 & G_2 are isomorphic if

- (i) Number of vertices are same.
- (ii) Number of edges are same.
- (iii) An equal number of vertices with given degree.
- (iv) Vertex correspondance & edge correspondance valid.

Handwritten notes on the left:

- $u=3$
- $v=3$
- $w=2$
- $x=3$
- $y=1$
- $p=2$
- $q=3$
- $r=3$
- $s=3$
- $t=1$
- $y-f$
- $w-p$
- $v-s$
- $u-x$
- $x-y$

Example :

(i) Number of vertices are same.
(ii) Number of edges are same.
(iii) An equal number of vertices with given degree.
(iv)

$\Rightarrow$ Both are isomorphic.

But they are not isomorphic, because

- (i) Number of vertices and edges are same.
- (ii) Sequence of degree of vertices are same.
- (iii) Mapping of vertices are not same.

Handwritten notes on the left:

- $u_1=1$
- $u_2=2$
- $u_3=2$
- $u_4=1$
- $u_5=2$
- $u_6=1$
- $v_1=1$
- $v_2=2$
- $v_3=3$
- $v_4=1$
- $v_5=2$
- $v_6=1$

1 Show that the graph given below are not isomorphic.

Handwritten notes below the graphs:

- u_1-v_1
- u_2-v_2
- u_3-v_3
- u_4-v_4
- u_5-v_5
- u_6-v_6

(1) **Length of the walk** : The number of edges is called length of the walk.

(2) **Closed & open walk** : A walk is said to be closed if its origin & terminous vertex ($v_0 = v_n$) is equal otherwise it is called open walk.

Example : $fe_{11}ge_{10}ae_1f$ is closed walk.

Trail : Any walk having different edges is called trail.

Circuit : A closed trail is called circuit.

Example : $ae_1fe_{11}ge_9be_2a$

Path : A walk is called path if all vertices are not repeated.

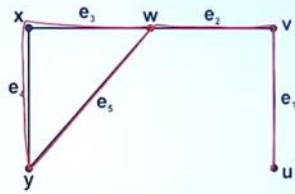
Cycle : A closed path is called cycle.

Example : $ae_{10}ge_7de_8ee_{13}fe_4a$

EULERIAN PATH

A path in a graph is said to be an Eulerian path if it traverses each edge in the graph once and only once.

Example :

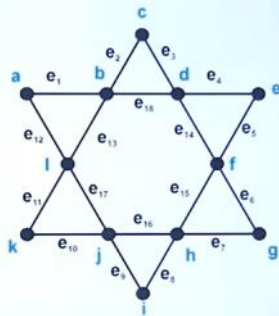


then $ue_1ve_2we_3xe_4ye_5w$

EULERIAN CIRCUIT

A circuit in a graph is said to be an Eulerian circuit if it traverses each edge in the graph once & only once.

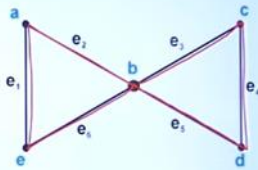
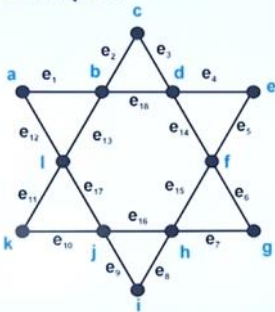
Example :



EULERIAN GRAPH

A connected graph which contain an Eulerian circuit is called Eulerian graph..

Example :



HAMILTONIAN PATH

A path which contains every vertex of a graph G exactly once is called a Hamiltonian Path.

Example :

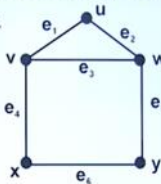


then $ue_1ve_2we_3xe_4y$

HAMILTONIAN CIRCUIT

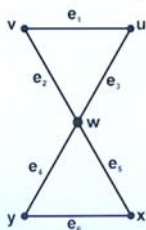
A circuit that passes through each of the vertices in a graph G exactly once except the starting vertex & end vertex is called a Hamiltonian circuit.

Example :



then
 $ue_1ve_2we_3xe_4ye_5u$

but

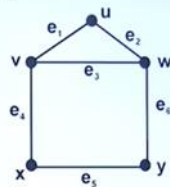


then
 $ue_1ve_2we_3xe_4ye_5u$
is not a Hamiltonian circuit
because vertex w repeats.

HAMILTONIAN GRAPH

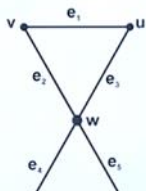
A connected graph which contains a Hamiltonian circuit is called a Hamiltonian Graph.

Example :



is a Hamiltonian Graph

Example :

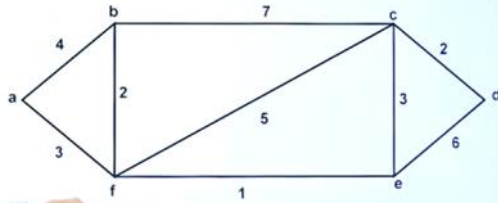


is not a Hamiltonian Graph.

WEIGHTED GRAPH

A graph is called weighted graph if a non-negative integer $W(e)$ associate to each edge and this $W(e)$ is a weight of corresponding edge.

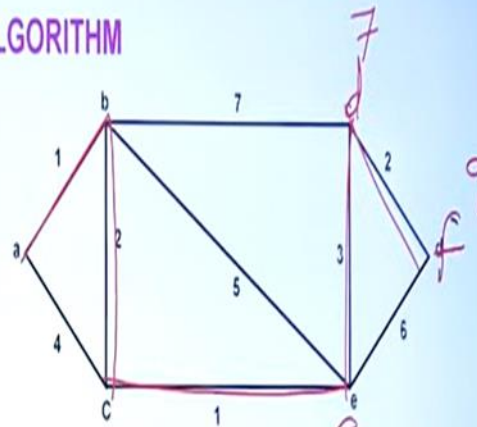
Example :



SHORTEST PATH IN WEIGHTED GRAPH

① a b c d e f DIJKSTRA'S ALGORITHM
 $\infty \infty \infty \infty$

Example :



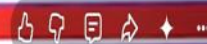
	a	b	c	d	e	f
a	0	∞	∞	∞	∞	∞
b	0	1	4	∞	∞	∞
c	0	1	3	8	6	∞
d	0	1	3	8	4	∞
e	0	1	3	7	4	10
f	0	1	3	7	4	9

a b c d e f



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Questions at : www.mathscare.com

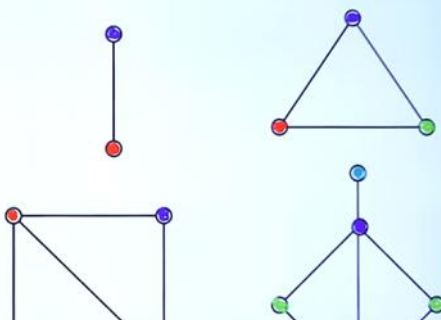


6:51 / 22:23 Dijkstra algorithm >

GRAPH COLORING

Painting all the vertices of a graph with colors such that no two adjacent vertices have the same color is called coloring of graph.

Example :



① a b c d e f

10 5 3 ∞ 2

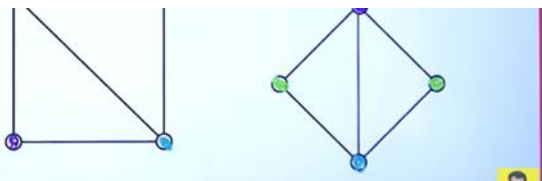
0 5 3 6

0 4 3 5

0 4 3 5

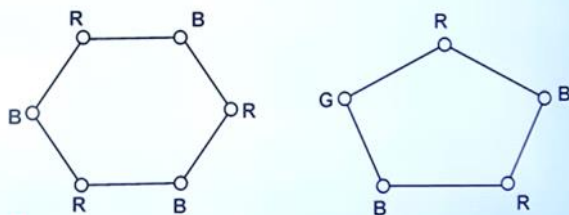
1 Compute shortest path for the following graph between a and d.

CHROMATIC NUMBER



Note :

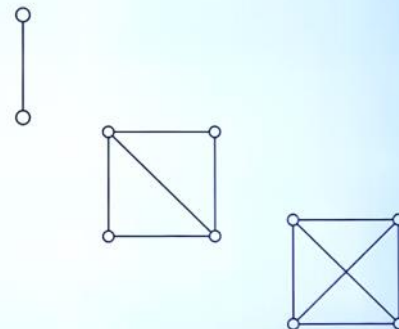
1. The chromatic number of graph G is denoted by $\chi(G)$.
2. If $\chi(G) = k$, then the graph is called k -chromatic.
3. Chromatic number of null graph is 1.
4. Chromatic number of complete graph K_n of n vertices is n .
5. If a graph is circuit with n vertices then
 - (i) It is 2-chromatic if n is even.
 - (ii) It is 3-chromatic if n is odd.



CHROMATIC NUMBER

The least number of colors required for coloring of a graph G is called its chromatic number.

Example :



TREE AND ITS PROPERTIES

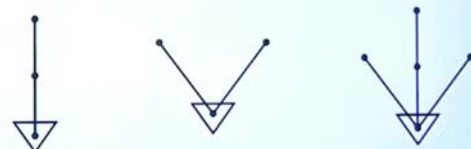
TREE : A tree is a connected graph without any loop or circuits.

Example :



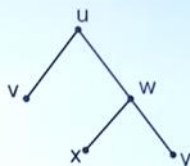
ROOTED TREE : A rooted tree is a tree in which one vertex is root.

Example :



PENDENT VERTEX IN TREE : A vertex of degree one is called pendent vertex of tree.

Example :



PATH LENGTH OF TREE : A path length of a tree is defined as the sum of edges from the root of all pendent vertices.

Example :



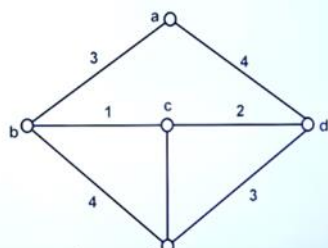
ALGORITHMS FOR MINIMAL SPANNING TREE

(1) KRUSHKAL'S ALGORITHM

Working Rule :

- (i) Choose an edge of minimal weight.
- (ii) At each step, choose the edge whose inclusion will not create a circuit.
- (iii) If G has n vertices, stop after $(n - 1)$ edges.

Example :

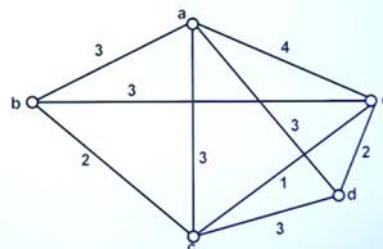


(2) PRIM'S ALGORITHM

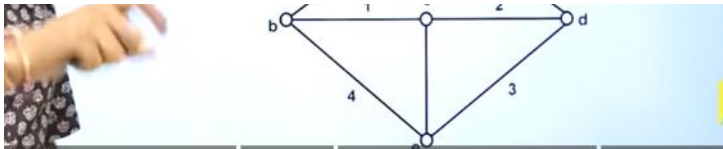
Working Rule :

- (i) Select any vertex & choose the edge and smallest weight from G .
- (ii) At each stage, choose the edge of smallest weight joining a vertex already included to vertex not yet included.
- (iii) Continue until all vertices are included.

Example :



Max degree of Any vertex in simple graph is $(n-1)$.
 Acc. Handshaking lemma
 $\sum \text{deg}(v) = 2E$



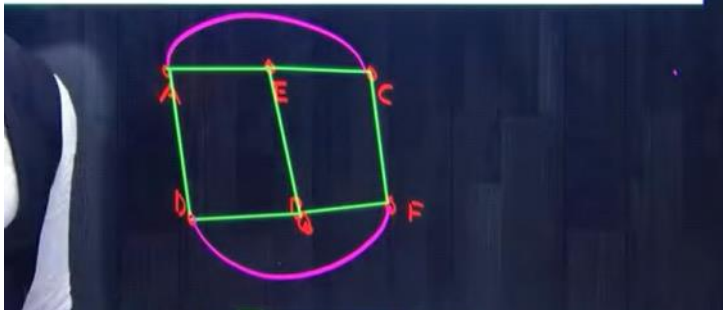
Planar Graph:

A graph is said to be planar if it can be drawn on a plane in a way that no edges cross one another.

Draw a planar representation of graphs given below if possible.

i) ii)

Fig. 5.15.7



Find the minimum spanning tree for the graph given in the following figure using Prim's algorithm.

By Prim's algorithm.

Step 1:

Step 2:

Step 3: b connected to d. $MST = \{(a,b), (b,d)\}$

Step 4: d connected to f. $MST = \{(a,b), (b,d), (d,f)\}$

Step 5:

Max degree of Any vertex in simple graph is $(n-1)$.

Acc. Handshaking lemma

$$\sum_{i=1}^n d(v_i) = 2 \times e$$

$$d(v_1) + d(v_2) + d(v_3) + \dots + d(v_n) = 2 \times e$$

$$(n-1) + (n-1) + (n-1) + \dots = 2 \times e$$

$$n(n-1) = 2 \times e$$

$$\frac{n(n-1)}{2} = e$$

Show that in a connected planar graph with 6 vertices, 12 edges each of region is bounded by 3 edges.

$V = 6$
 $E = 12$
 $R = ?$

Euler's Formula.

$$V - E + R = 2$$

$$6 - 12 + R = 2$$

$$R = 8$$

Show that in a connected planar graph with 6 vertices, 12 edges each of region is bounded by 3 edges.

$V = 6$
 $E = 12$
 $R = ?$

Total Edges = $12 \times 2 = 24$

No. of Edges Required for each Region = $\frac{24}{3} = 8$

Euler's Formula.

$$V - E + R = 2$$

$$6 - 12 + R = 2$$

$$-6 + R = 2$$

$$R = 2 + 6$$

$$R = 8$$

Most Probable Questions (Chapter-wise)

Chapter 6: Graph Theory

- **Graph Properties:**
 - Sum of degrees is even (2025) ★
 - Complete graph K_n has $n(n-1)/2$ edges (2025) ★
 - Simple graph has two vertices of same degree (2025).
- **Connectivity & Walks:**
 - Euler path/circuit condition for $K_{m,n}$ (2025). ★
 - Define Eulerian/Hamiltonian walks (2023).
- **Graph Coloring:**
 - Chromatic number of planar graph (≤ 4) (2025, 2024).
 - Every tree is 2-chromatic / bipartite (2018, 2023). ★
 - Chromatic polynomial of K_n or small graph (2017, 2018).
- **Trees:**
 - Tree with n vertices has $n-1$ edges (connected, no cycles) (2024).
 - Max edges in graph with n vertices & k components (2024).
 - At most one perfect matching in a tree (2017).

Most Probable Questions (Chapter-wise)

Chapter 4: Propositional Logic

- **Truth Tables:** Check logical equivalence of two statements (2025) ★, Prove tautology (2019).
- **Laws of Logic:** De Morgan's Law (both logic & set forms) (2024, 2023) ★, Absorption Law.
- **Normal Forms:** Find DNF and CNF without truth table (2024, 2018). ★
- **Validity of Arguments:** "Program runs / submit project / computer crashes" type (2019, 2018). ★
- **Quantifiers:** "Nobody likes everybody" translation (2017), Symbolization of statements.

Most Probable Questions (Chapter-wise)

Chapter 2: Principles of Induction & Number Theory

Prove by Induction: $n^2 + n$ is even (2023) ★, $3n < n!$ for $n \geq 6$ (2024) ★, Sum of first n natural numbers.

State Theorem: Fundamental Theorem of Arithmetic ★, Division Algorithm ★.

GCD Application: Show $\gcd(a^2 - b^2, a^2 + b^2) = 1$ or 2 (2019). ★

Modular Arithmetic: Prove $7 \mid (2^{(5n+3)} + 5^{(2n+3)})$ (2019). ★

Euclidean Algorithm: Find integers u, v such that $\gcd(272, 119) = 272u + 119v$ (2018, 2017). ★

Chapter 3: Combinatorics ★★ ★ (Most predictable)

- Pigeonhole Principle (PHP):
 - *Classic*: Minimum socks to pull for a match (2024) ★
 - *Proof*: In a party, two people shake same number of hands (2024). ★
 - *Generalized PHP*: "At least 4 born in same month" (2017).
- Inclusion-Exclusion (PIE):
 - *3-set problem*: Students in ML, DS, AI – find at least one, exactly one, none (2025). ★
 - *2-set problem*: Swimming/Jogging (2018).
 - *Number of integers* divisible by 2,3,5 between 1 and 1000 (2017, 2019).
- Permutation & Combination:
 - Arrangements with conditions (e.g., specific people together) (2025). ★
 - Committee formation with "at least 1 woman" (2025).
 - Non-negative integer solutions: $x_1 + x_2 + x_3 + x_4 = 20$ (2023, 2018). ★
 - Probability: A speaks before B (2023).

Chapter-Wise Focus Suggestions

1. Set Theory, Relations, Functions (Easy scoring)

- **Memorize definitions**: Power set, Cartesian product, Domain/Range, types of relations.
- **Practice**: Symmetric difference (Δ) and complement problems.
- **Key PYQ**: $A \Delta B$ calculation, number of reflexive relations = $2^{n^2 - n}$.

2. Number Theory & Induction (Medium – proof heavy)

- **Must practice**: Proof by induction (at least 3 varieties: divisibility, inequality, summation).
- **Must know**: Euclidean Algorithm steps, GCD properties, Modular arithmetic rules.
- **Key PYQ**: $7 \mid (2^{5n+3} + 5^{2n+3})$ appears in 2019 & similar.

Chapter-Wise Focus Suggestions

3. Combinatorics (High scoring – problems are mechanical)

- **PHP**: Memorize classic statements (socks, birthdays, handshake).
- **PIE**: Practice 3-set Venn diagram problems thoroughly. Very high chance of 2025's ML/DS/AI problem repeating with different numbers.

Chapter-Wise Focus Suggestions

3. Combinatorics (High scoring – problems are mechanical)

- **PHP:** Memorize classic statements (socks, birthdays, handshake).
- **PIE:** Practice 3-set Venn diagram problems thoroughly. **Very high chance** of 2025's ML/DS/AI problem repeating with different numbers.
- **Combinations:** "Stars and bars" formula for non-negative solutions. $C(n+r-1, r-1)$. ★
- **Key PYQ:** 2025's committee & PIE problem is a **goldmine**.

4. Propositional Logic (Conceptual + problems)

- **Must know:** Truth tables for all connectives, logical equivalence laws (De Morgan, Distributive, Implication).
- **Practice:** Tautology proof using laws (no truth table). Normal form conversion (DNF/CNF) – often in Group-C.
- **Key PYQ:** 2024's DNF/CNF without truth table, 2025's equivalence with truth table.



Chapter-Wise Focus Suggestions

5. Algebraic Structures (Abstract but pattern exists)

- **Group:** Focus on cyclic groups, generators, subgroup tests. $U(7) = \{1, 6\} \pmod{7}$ is a classic cyclic group example.
- **Ring:** Practice commutativity proofs given special condition (like $x^3 = x$).
- **Lattice:** Hasse diagram drawing, checking distributivity, complement.
- **Key PYQ:** 2024's group of order 15 is cyclic (uses Sylow or Cauchy – memorize the proof).

6. Graph Theory (Visual + formulas)

- **Must memorize:** Handshaking Lemma, K_n edges formula, Tree properties, Euler path conditions.
- **Coloring:** Chromatic number of bipartite = 2, complete = n , cycle = 2/3. Chromatic polynomial of simple graphs.
- **Algorithms:** Dijkstra's (step-by-step) – 2024 had it, likely to repeat.
- **Key PYQ:** 2025's degree sum & K_n edges proof, 2024's Euler path in $K_{\{m, n\}}$.

Repeatedly Asked / High Probability Questions (Must Prepare)



Prove sum of degrees of all vertices in a graph is even.

Prove a complete graph K_n has $n(n-1)/2$ edges.

Find GCD using Euclidean Algorithm & express as linear combination.

Inclusion-Exclusion problem with 3 sets (students in courses).

Show $G = \{1, 2, 3, 4, 5, 6\}$ is a cyclic group under $\times \text{ mod } 7$.

Pigeonhole Principle: socks / handshake / birthday.

Prove by induction: $n^2 + n$ is even / $3n < n!$.



Repeatedly Asked / High Probability Questions (Must Prepare)



Truth table to prove logical equivalence or tautology.

Find DNF & CNF of a given logical expression.

Check validity of argument using rules of inference.

Every tree is bipartite / 2-chromatic.

Non-negative integer solutions to

$$x_1 + x_2 + \dots = n.$$

Final Exam Strategy

Group A

Focus on definitions (Cantor, Symmetric difference, Abelian group, Chromatic number, Spanning tree, Tautology). Each correct = easy 1 mark.

Group B

Choose PHP or small PIE or Induction or simple graph proof. Avoid complex algebra unless confident.

Group C

- One from **Graph Theory** (degree sum + edges proof OR coloring)
- One from **Combinatorics** (PIE with 3 sets) ★
- One from **Group Theory** (cyclic group proof) OR **Logic** (DNF/CNF or argument validity)